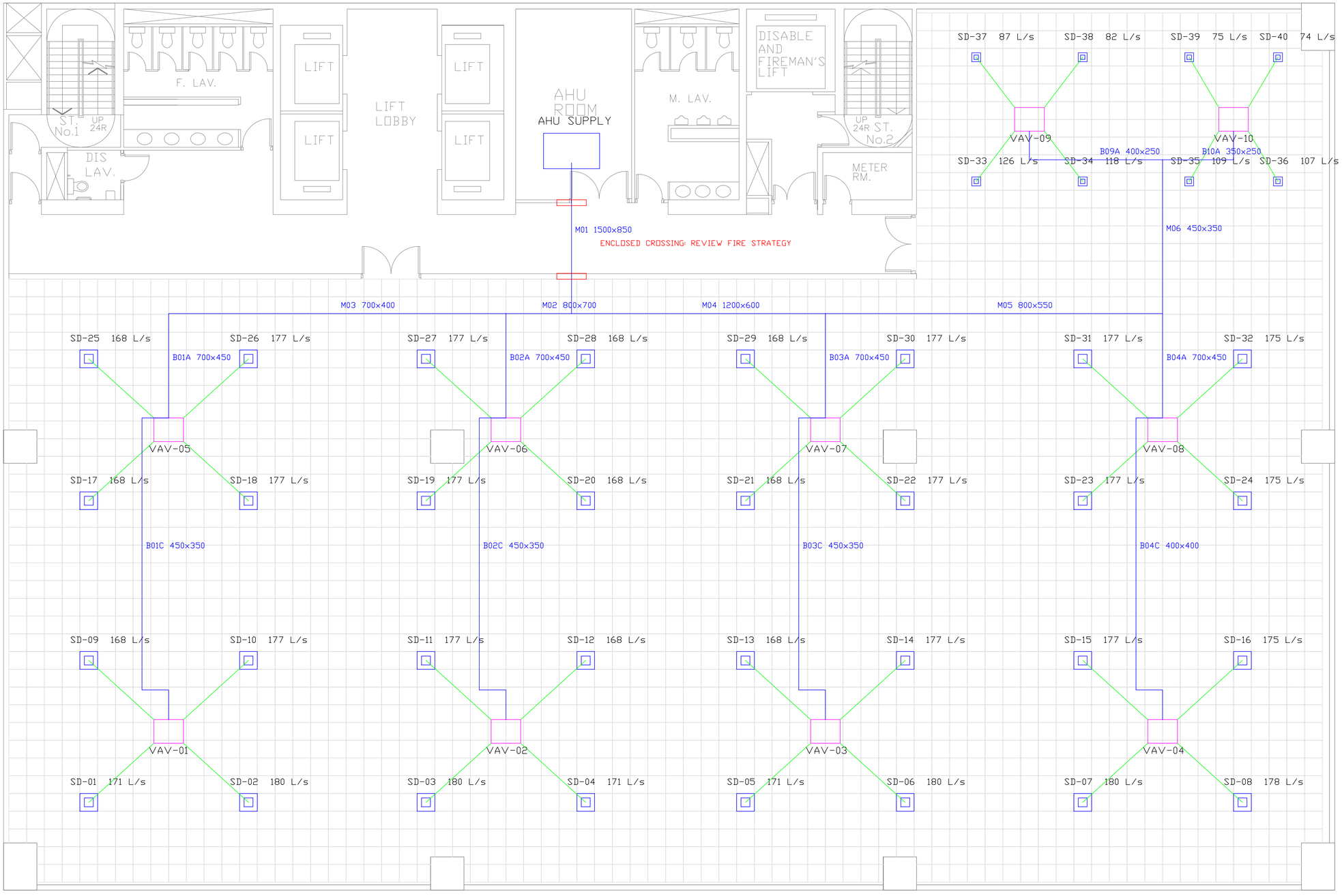


BSE3227 / HVACR 1

HVAC REVIEW LAYOUT v01

PRELIMINARY - NOT FOR SUBMISSION

40 SUPPLY DIFFUSERS / 10 VAV BOXES

4 outlets per VAV box

Printed office area: 1022 m2

Assumed total load: 120 W/m2

Assumed SHR: 0.75; room/supply 24/12 C

Peak supply: 6349.4 L/s

Minimum supply: 2539.8 L/s

BLUE: rigid supply duct / diffuser

GREEN: flexible connection

MAGENTA: VAV (schematic envelope)

All flex routes include 0.50 m allowance.

Maximum calculated flex: 3.474 m

Flex limit: 4 m (FSD 2023 cl. 7.2)

Assignment limit: 5 m

Duct sizes: width x height, mm

Main velocity <= 5.0 m/s

Branch velocity <= 4.5 m/s

Straight rigid friction <= 0.8 Pa/m

Rectangular aspect ratio <= 4:1

ADPI REVIEW REMAINS OPEN:

7 outlets fail initial course screen.

Manufacturer pattern classification

requires resolution before acceptance.

Ceiling / beam clearances assumed;

coordinate before construction.

AHU corridor crossing requires fire

enclosure / boundary protection review.

Source architecture retained.

See CSVs for all 30 rigid sections.

TYPICAL FLOOR / SUPPLY AIR SINGLE-LINE PLAN

HV-01 | A2 1:100 | 2026-11-08 | DESIGN REVIEW v01